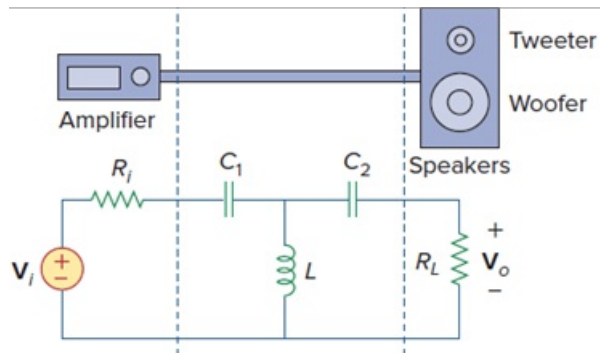


Problem

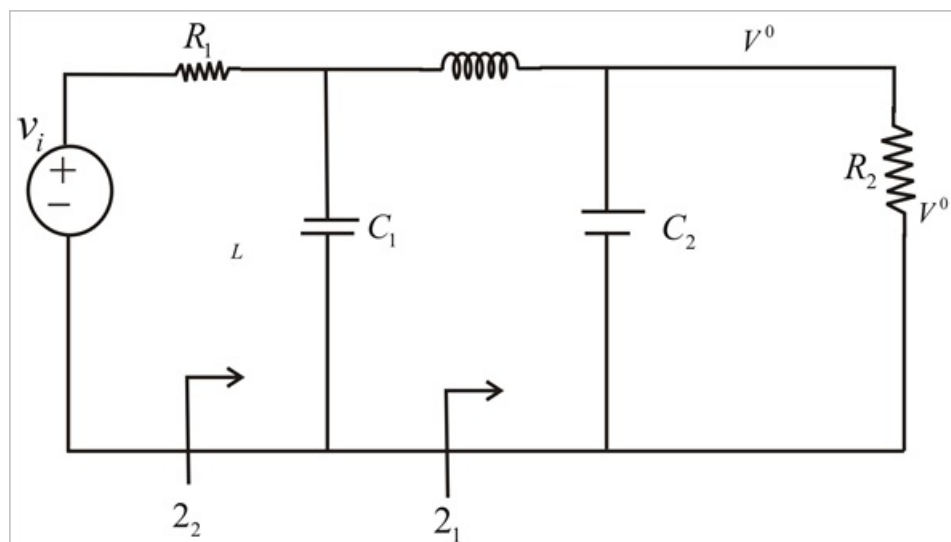
The crossover circuit in Fig. 14.109 is a high-pass filter that is connected to a tweeter. Determine the transfer function $H(\omega) = V_o(\omega)/V_i(\omega)$.

Figure 14.109:



Step-by-step solution

Step 1 of 2



Step 2 of 2

$$Z = SL \parallel \left[RL + \frac{1}{SC_2} \right] = \frac{SL \left(R_L + \frac{1}{SC_2} \right)}{R_L + SL + \frac{1}{SC_2}} \quad , \quad S = j\omega$$

$$V_1 = \frac{Z}{Z + R_i + \frac{1}{SC_1}} V_i$$

$$V_o = \frac{R_L}{R_L + \frac{1}{SC_2}} V_1 = \frac{R_L}{R_L + \frac{1}{SC_2}} \cdot \frac{Z}{Z + R_i + \frac{1}{SC_1}} V_i.$$

$$H(\omega) = \frac{V_o}{V_i} = \frac{R_i}{R_L + \frac{1}{SC_2}} \cdot \frac{1}{SL} \frac{SL \left(R_L + \frac{1}{SC_2} \right)}{\left(R_L + \frac{1}{SC_2} \right) + \left(R_i + \frac{1}{SC_1} \right) \left(R_L + SL + \frac{1}{SC_2} \right)}$$

$$H(\omega) = \frac{S^3 LR_L C_1 C_2}{(SR_i C_1 + 1)(S^2 LC_2 + SR_L C_2 + 1) + S^2 LC_1 (SR_L C_2 + 1)}$$

where $S = j\omega$